

Alex Granada

Systems Engineering Student | Software Developer

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Profile

Systems Engineering student interested in software development, low-level programming, and understanding how systems work beneath their abstractions. I enjoy building projects independently, learning unfamiliar technologies through experimentation, and combining software with other areas such as electronics, graphics, and physical fabrication.

Education

Systems Engineering

2024 – Present

Caldas University

Relevant areas: programming, data structures and algorithms, computer architecture, databases, computer networks, electronics, mathematics, and software development.

Selected Projects

Chipra FM 500KHz

2026

C, WebAssembly, JavaScript, HTML/CSS

Simulated FM radio encoder/decoder running in the browser. Implemented the signal-processing core in C and compiled it to WebAssembly, with a browser-based interface for experimentation and visualization.

[Demo](#) [Source](#)

YY-font

2025

C++

Modular font parser and rasterizer implemented primarily as a single-header C++ library. Developed to explore font file formats, binary data, parsing, and low-level graphics.

[Source](#)

Correlones de Canaguay

2025

Python

Small game developed around data structures and algorithms, featuring procedurally created roads and intentionally chaotic vehicle behavior.

[Source](#)

Carechucho

2025

Python

Short experimental game created as a practical exercise in data structures and algorithms.

[Source](#)

Personal Projects & Fabrication

Ongoing

Software, electronics, sewing, 3D design

Independent projects spanning plush design and fabrication, electronics experiments, interactive web projects, graphics, and other small-scale technical and artistic experiments.

Technical Skills

Programming: C, C++, Python, JavaScript, TypeScript

Web: HTML, CSS, JavaScript, TypeScript, Node.js, React, Angular

Low-level / Systems: WebAssembly, assembly, ABIs, binary file formats, parsing, memory management, C/C++ tooling

Tools: Git, CMake, LaTeX, GitHub, VS Code

Other: Godot, Unity, Blender, Inkscape, MuseScore

Additional Interests

Low-level software, programming language implementation, computer graphics, electronics, embedded systems, digital fabrication, plush design, music, and interactive art.

Languages

Spanish Native

English Intermediate (empirical)

Online

GitHub github.com/Alex461538

Personal website alex461538.github.io/Alex461538/

YouTube youtube.com/@pianitas38